SAVEETHA SCHOOL OF ENGINEERING
SAVEETHA INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES
DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

COURSE NAME: STATISTICS WITH R PROGRAMMING **COURSE CODE: ITA04**

COURSE OUTCOME COVERED

CO1: *Apply R Programming fundamentals including data structures, vectors, matrices, arrays and class types to perform mathematical operations and manage data effectively. (BL: 3)*

ASSESSMENT TOOL 1: Pseudocode Writing Task (Algorithm Design) Level2

Assessment Tool Weightage: 30%

QUESTIONS: 5 **Total Marks: 50** **Duration: 1 Hour**

Code of Conduct:

I S. Harish *(Name)* 192521003 *(Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

S. Harish

Signature of the student:

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary (5 Marks)	Level 3 – Proficient (4 Marks)	Level 2 – Developing (2–3 Marks)	Level 1 – Beginning (0–1 Mark)	Q1 (10)	Q2 (10)	Q3 (10)	Q4 (10)	Q5 (10)	Total Marks (50)
Conceptual Understanding	30	Demonstrates clear and complete understanding of the concept	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor or incorrect understanding						
Accuracy of Answer	25	Completely correct answer with no errors	Mostly correct with minor errors	Partially correct with noticeable errors	Incorrect or irrelevant answer						
Application of Knowledge	20	Correctly applies concepts to solve the question/problem	Applies concepts with minor mistakes	Limited or partially correct application	No or incorrect application						
Completeness of Response	15	Covers all required parts of the question	Covers most parts with minor omissions	Covers only some parts	Incomplete or missing response						
Clarity & Presentation	10	Clear, concise, and well-organized answer	Understandable with minor issues	Somewhat unclear or poorly structured	Difficult to understand						
					Total per Question						

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1.) write a pseudo code for a employee salary processing ?

program:

```
BEGIN
    Create data frame Employee with:
        ID, Name, Department, Salary
    create logical vector:
        Eligible = Salary > 50000
    Create list EligibleEmployees
        Store Employee details where IT is TRUE
    Display Elibible employees
    count = Number of eligible employees
    Display count
    END
```

OUTPUT

Eligible Employee

101	hari	IT	100000
103	priya	CSE	75000

Number of eligible employee : 2

2.) write a pseudo code for inventory managment?

Program:

START

create data frame Inventory with:

Item name, quantity, unit price

Calculate Total value = Quantity * unit price

create logical vector:

Low Stock = Quantity < minimum level

Store low-Stock items in a list

Display low Stake items

END

OUTPUT

Item	Quantity	Total value
		30
Pen	3	
Note book	20	1000

Low Stock Items

Pen.

3.) write a pseudocode for Examination Result Processing?

Program:

```
STRAT
    Create vectors StudentNames & marks
    Define function AssignGrade(marks):
        IF marks >=90, Grade = "A"
        ELSE IF marks >=75, Grade = "B"
        ELSE IF marks >=50, Grade = "C"
        ELSE Grade = "F"
        RETURN GRade
    APply Assign Grade to all marks
    Store grades as a factor
    Create data frame with:
        Name, marks, Grade
    count Students in each category
    Display data frame and grade counts
END
```

OUTPUT

Name	mark	Grade
Rani	80	A
Hari	95	S
bala	90	A

Grade counts:

A = 2

S = 1

4.) write a pseudo code for weather data processing

Program

BEGIN

create vector Temperatures

create logical vector:

Hot days = Temperature > 35

count the number of TRUE in Hot days

Store Temperature, Hot days as count in list

Display the list

END

OUTPUT

Temperatures : 32, 68, 38, 30, 37

Hot days : FALSE, TRUE, TRUE, FALSE, TRUE

number of hot days : 3

5.) Write a pseudo code for Library Book Analysis?

Program

START

Create data frame BOOK with:

BOOK ID, Title, Copies

Create logical vector:

Replenish = Copies < 5

Store Title where Replenish IS TRUE

in a character vector

Display books requiring

END

OUTPUT

Book requiring replenishment:

Python Basics

DBMS

Computer Networks